

NANYANG JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION

Higher 1

MATHEMATICS

8864/01

Paper 1

11 September 2012

3 hours

Additional Materials: Cover sheet
 Answer papers
 Graph paper
 List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your name and class on every script you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagram or graph.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of 5 printed pages.

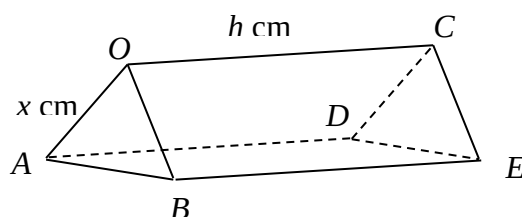


NANYANG JUNIOR COLLEGE
Internal Examinations

Section A: Pure Mathematics [35 marks]

- 1 A point $P(x, y)$ moves along a curve given by the equation $y = e^{2x}$. Given that y is increasing at a constant rate of 3 units per second, find the exact rate of change of x at the instant when $x = \ln 3$. [3]
- 2 Find algebraically, the set of values of k for which the expression $kx^2 + 4x + (2k - 7)$ is always positive for all real values of x . [4]
- 3 (i) Find $\int \frac{1}{\sqrt{1-2x}} dx$. [2]
- (ii) Without using a calculator, find the exact value of $\int_3^8 \frac{1}{1-3x} dx$. [3]
- 4 (i) On a single diagram, sketch the graphs of $y = \ln(5-2x)$ and $y = \sqrt{x} - 3$, stating clearly equation(s) of asymptote, if any. [2]
- (ii) Find the coordinates of the point(s) of intersection of $y = \ln(5-2x)$ and $y = \sqrt{x} - 3$, giving your answer correct to 4 decimal places. [1]
- (iii) Write down as an integral an expression for the area of the region bounded by $y = \ln(5-2x)$, $y = \sqrt{x} - 3$ and the line $x = 0$. Evaluate this integral, giving your answer correct to 3 decimal places. [2]
- 5 The curve C is given by the equation $y = \frac{8}{3(2x-1)^3} + \ln(2x-1)$, where $x > \frac{1}{2}$. The point P lies on the curve and has x -coordinate 1.
- (i) Show that C has exactly one stationary point. [4]
- (ii) Find the equation of the tangent to the curve at the point P , giving your answer in the form $y = ax + b$, where a and b are to be determined exactly. [2]
- (iii) The tangent to C at P meets the y -axis at Q . Find the exact area of triangle OPQ . [2]

- 6 The diagram below shows a solid prism $OABCDE$ such that the cross-section OAB is an equilateral triangle with side x cm and $OC = AD = BE = h$ cm. The prism has a fixed volume of 1000 cm^3 .



- (i) Show that the total surface area, $S \text{ cm}^2$, of the prism is given by

$$S = \frac{\sqrt{3}}{2}x^2 + \frac{4000\sqrt{3}}{x} \quad [2]$$

- (ii) Use differentiation to determine the exact value of x for which S is a minimum. State the corresponding minimum value of S , giving your answer correct to 1 decimal place, and explain how you can tell that it is a minimum rather than a maximum. [4]

- (iii) Given that $10 \leq x \leq 20$, find

- (a) the largest value that S can take, and
 (b) the range of values of x where $S \geq 670$.

You need to show clearly how you obtain your answer. [4]

Section B: Statistics [60 marks]

- 7 An estate developer wishes to get the opinions of the residents in a particular estate on whether they would like to have roller-blading facilities in a new park to be constructed.

- (i) Describe how the developer could obtain a quota sample of size 50. [2]
 (ii) The residents in the estate consist of 3 different age groups: less than 15 years old, 15 to 25 years old and 26 years old and above. If the ratio of the residents in the 3 different age groups is $3 : 2 : 3$, how should the developer select a stratified sample of 80 residents? [2]

- 8 The events A and B are such that $P(A) = 0.3$, $P(B') = 0.35$ and $P(B|A) = 0.5$. Find the probability that

- (i) both A and B occur, [1]
 (ii) at least one of A and B occurs, [2]
 (iii) exactly one of A and B occurs. [2]

[Turn Over]

- 9** On a certain island, the probability that it will rain on any given day is 0.7, independently of any other day.
- (i)** Find the probability that, in a given week of 7 days, it will rain on at least 4 days, giving your answer correct to 5 decimal places. [2]
 - (ii)** Find the probability that, over a period of 20 weeks, there are more than 9 weeks but fewer than 15 weeks in which it will rain on at least 4 days per week. [3]
 - (iii)** Using a suitable approximation, find the probability that, over a period of 20 weeks, it will rain on at most 100 days. State the mean and variance of the approximation. [4]
- 10** A bag contains 8 red balls and 1 blue ball. A game is played by drawing 3 balls, one at a time, from the bag. Each time a red ball is drawn, it is replaced in the bag along with 2 additional red balls. Each time a blue ball is drawn, it is replaced into the bag.
- To win the grand prize, a player has to draw 3 blue balls. A player gets a consolation prize if he obtains exactly 2 blue balls in his three draws. Otherwise, he walks away empty-handed.
- Draw a tree diagram to represent the possible outcomes of the three draws. [1]
- Find the probability that
- (i)** a player wins a consolation prize, [2]
 - (ii)** a player wins a consolation prize, given that he drew a blue ball in his second draw. [3]
- A player plays the game three times. Find the probability that he wins the grand prize twice. [3]
- 11** A car company which manufactures sport cars has business dealings in various countries. The number of sport cars sold by the company in each quarter, x , and the corresponding quarterly profits, y in thousand dollars, are given in the table below.

The number of sport cars sold, x	16	15	18	23	22	17	21	24
Profits in thousands, y	300	270	340	410	390	290	360	410

- (i)** Give a sketch of the scatter diagram for the data, as shown on your calculator. [1]
- (ii)** Calculate the equation of the regression line of y on x , in the form $y = ax + b$, giving a and b in 4 decimal places. Draw this line on your scatter diagram. [2]
- (iii)** Find $\bar{x}$ and $\bar{y}$, and mark the point $(\bar{x}, \bar{y})$ on your scatter diagram. [2]
- (iv)** Calculate the product moment correlation coefficient, and comment on its value in relation to your scatter diagram. [2]
- (v)** Estimate the quarterly profit when 20 sport cars are sold in a quarter. Comment on the reliability of the estimate. [2]

- 12** An electronic device was advertised to be able to retain information for an average of 83 hours after external power to it has been cut off. Experiments were carried out to test this claim. The retention time in hours, X , was measured for 200 devices, and the result are summarised by

$$\sum(x-80) = 746.5 \text{ and } \sum(x-80)^2 = 22178.9$$

- (i)** Find unbiased estimates of the population mean and variance. [2]
- (ii)** Test, at the 5% significance level, whether the mean retention time is indeed 83 hours. [4]
- (iii)** Explain, in the context of part (ii), the meaning of “at the 5% significance level”. [1]
- (iv)** Experiments were carried out on a second sample of 200 devices and the mean retention time of this sample was obtained. Assuming that the retention time has a standard deviation of 11 hours, find the set of values of the mean retention time of this sample to conclude that the retention time was wrongly advertised at 5% level of significance. [4]
State, giving a reason, whether it is necessary to assume that the retention time has a normal distribution for the test to be valid. [1]

- 13** Ann travels to work every morning by bus. The waiting time for the bus is normally distributed with mean 11 minutes and standard deviation 3 minutes. The journey on the bus is normally distributed with mean 20 minutes and standard deviation 5 minutes. She will then take a 2-minutes walk from the bus-stop to her office.

It may be assumed that the waiting time for the bus, the journey on the bus and the walking time are all independent.

- (i)** Given two particular days, find the probability that she waits no more than 5 minutes for the bus on one day and more than 5 minutes for the bus on the other day. [3]
- (ii)** By finding the total time she takes to go to work on a morning, find
 - (a)** the probability that she takes a total of between 35 minutes to 40 minutes to travel to work on a particular morning, [3]
 - (b)** the smallest value of k such that the probability that the total time Ann takes to go to work on a morning exceeds k minutes is at most 0.8, giving your answer to the nearest minute. [3]

After office hours, Ann takes the same route to go home. The total time she takes to travel home from work on any day is 1.5 times of the total travelling time on that morning. Find the probability that, on any randomly chosen day, the sum of her travelling time to and from work is less than one hour. [3]

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